

CLASH OF THE **CLAWS**

RVCE_SnapTask
Point your camera.
Skip the typing.

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Problem Statement

Theme: On-device AI + Productivity + GenAI Research

Problem: Every photo of a business card, event flyer, receipt, or whiteboard demands manual re-entry. Your phone captures the data but never acts on it — that gap is entirely left to you.



Who faces it: Students, working professionals, and anyone who regularly photographs cards, posters, and receipts — essentially every smartphone user.

Why it matters: Billions of photos are taken daily. Not one of them automatically creates the calendar event, saves the contact, or logs the expense it contains. The information exists — it just never moves

Current Solution & Gaps

Existing solutions: Google Lens (searches the web), iOS Live Text (copies text to clipboard), Samsung Bixby Vision (translates / searches). All stop at recognition.

What's missing: None of them execute the action. They hand you text and expect you to do the rest.

Opportunity: The entire workflow from "I see something useful" to "it's saved" can be automated. No existing solution on Android does this end-to-end.

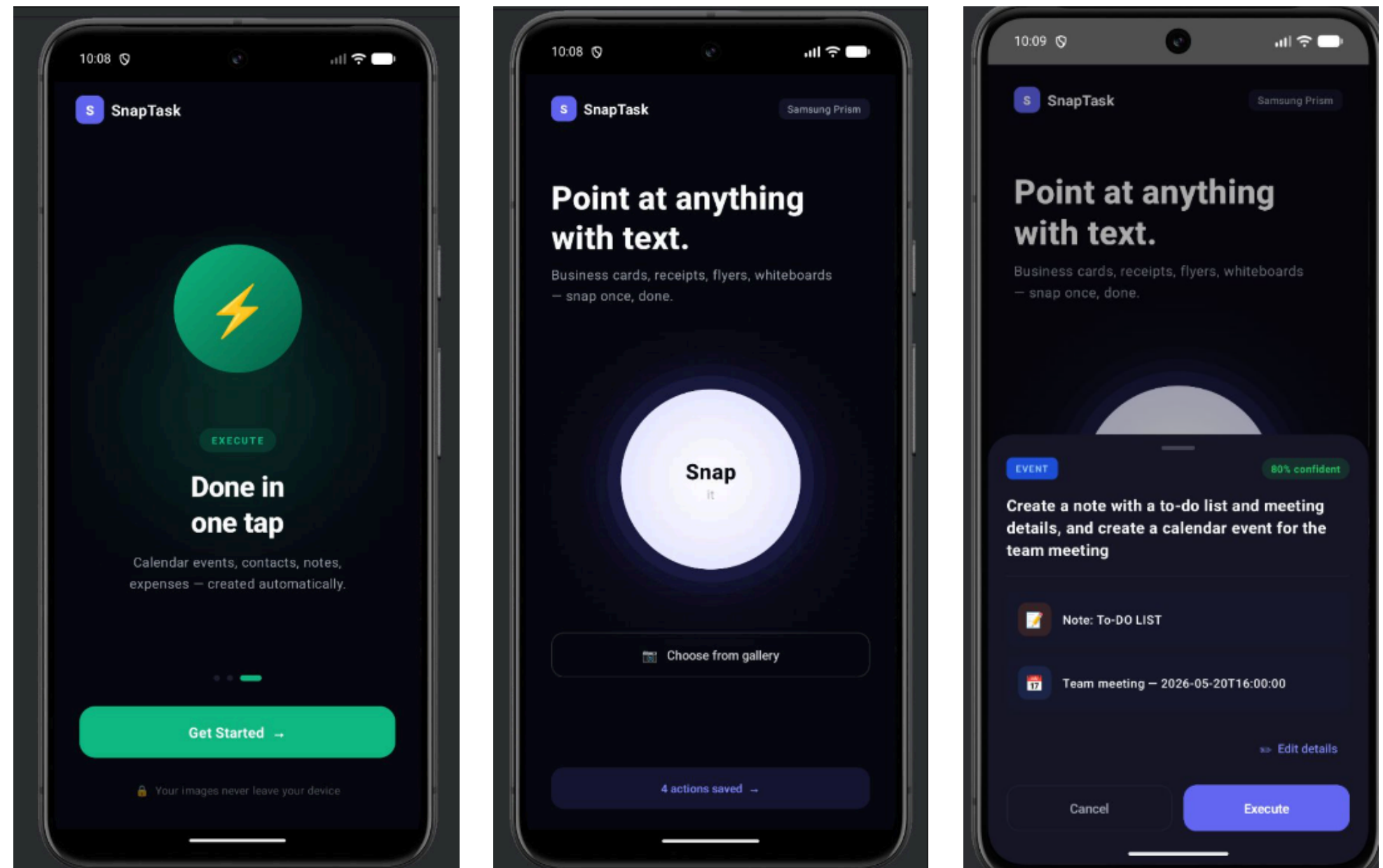
Your Solutions

SnapTask

An Android app that intercepts a photo, reads the text on-device, uses Gemini AI to classify intent, and executes the right Samsung API action — all with a single confirmation tap.

How It Works:

1. OCR processes image on-device
2. Groq Flash classifies intent
3. Extracts parameters for actions
4. Shows confirmation sheet
5. Executes via Samsung APIs



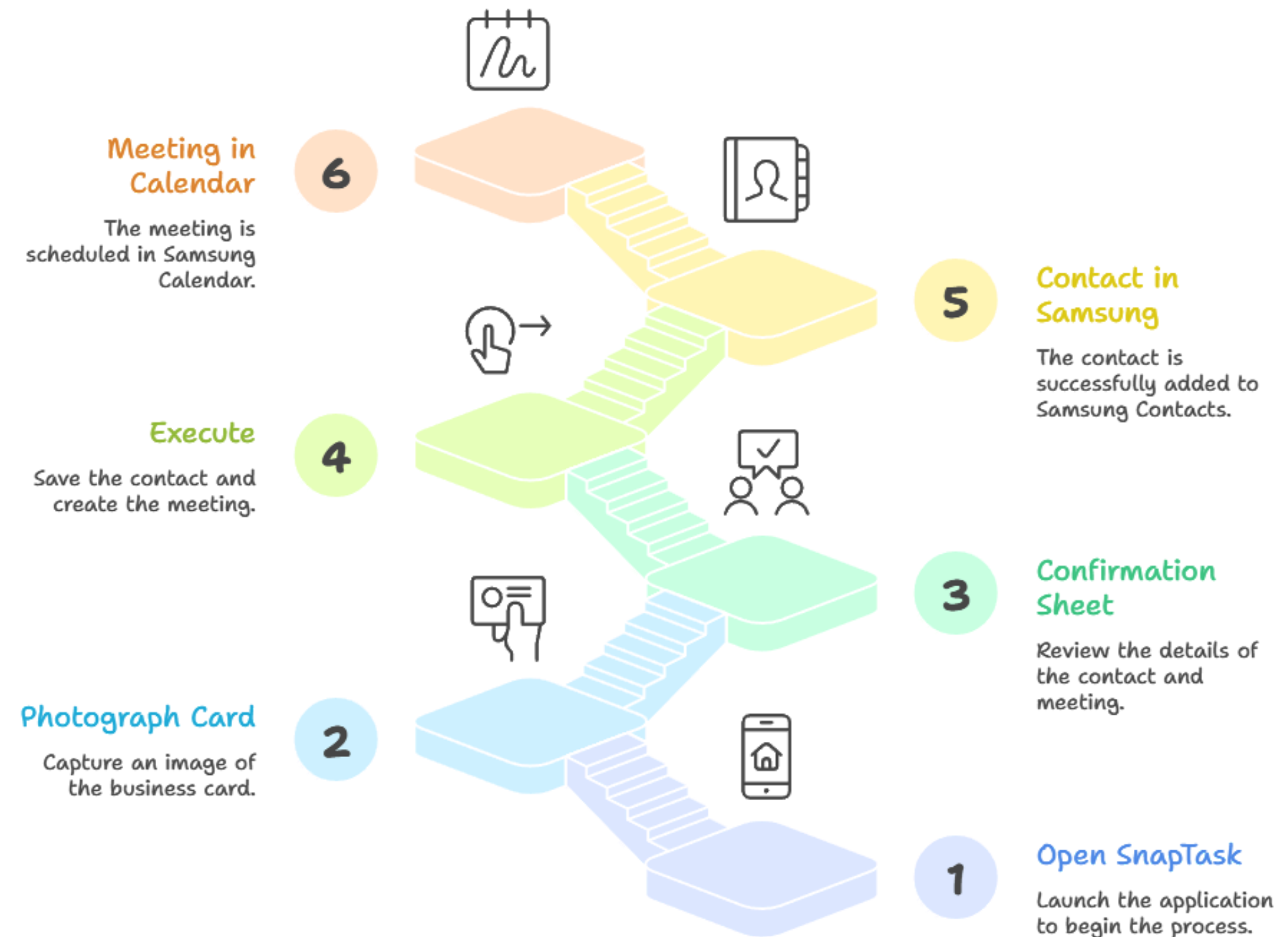
Demo/Product Walkthrough

Flow:

1. Open SnapTask → tap Snap it
2. Photograph a business card
3. Confirmation sheet slides up — shows "Savecontact: Jane Smith" + "Create follow-up meeting: May 15, 2:00 PM"
4. Tap Execute
5. Contact appears in Samsung Contacts. Meeting appears in Samsung Calendar.

Key moments:

- Multi-action in one snap (the "wow" moment)
- Amber confidence warning prompts review when AI is uncertain
- Edit details inline before executing — no blind saves
- history screen shows everything the app has done



Tech & Architecture

Android: Kotlin · Jetpack Compose · Material 3 · ML Kit OCR · Retrofit + OkHttp

**AI: Groq— structured JSON output, multi-action
prompt**

Server: Node.js OpenClaw gateway (runs on Mac, same Wi-Fi as phone)

**Samsung APIs: CalendarContract (Intent.ACTION_INSERT) · ContactsContract
· Samsung Notes SDK**

System flow:

Photo → ML Kit OCR (on-device) → rawText

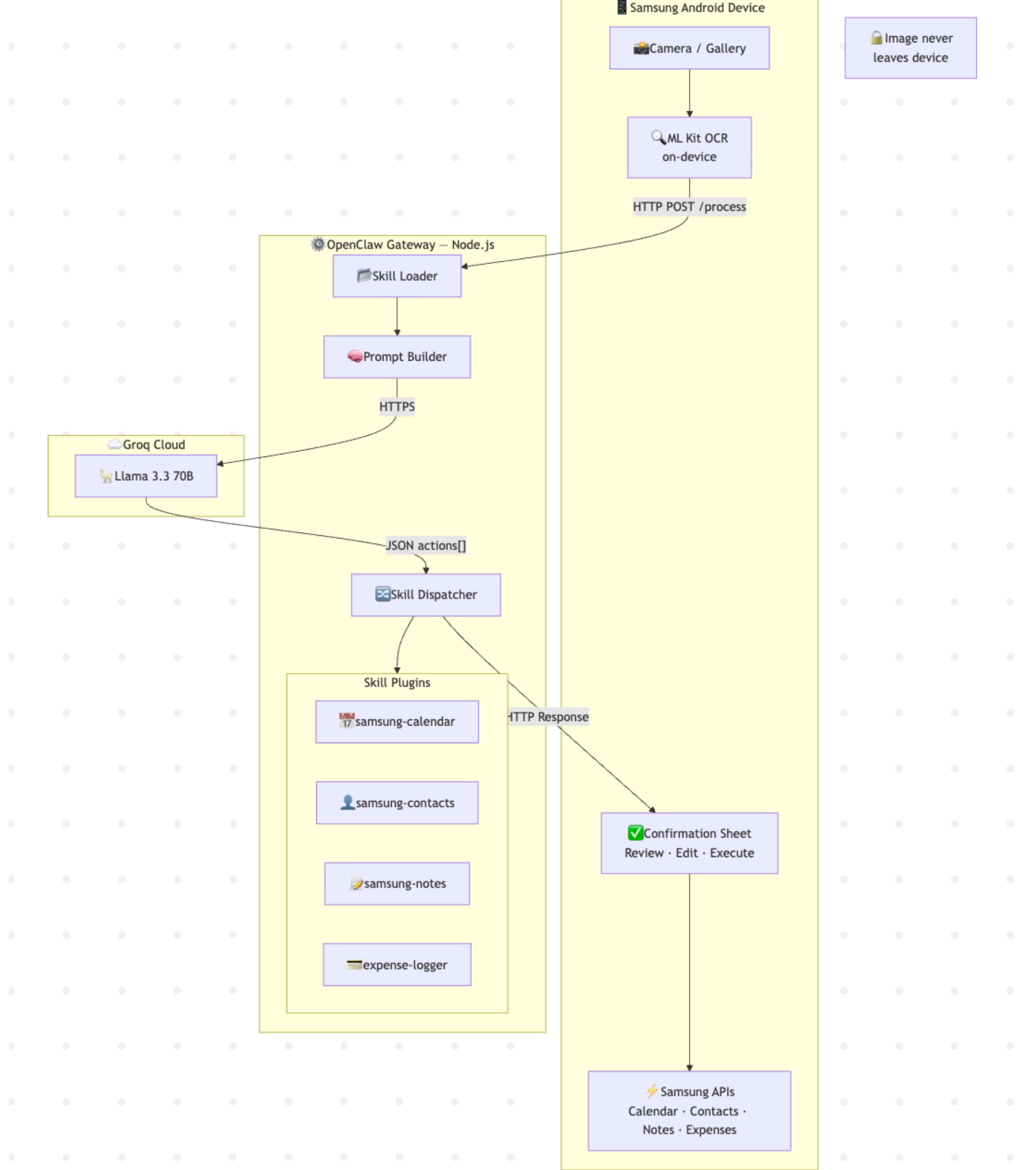
→ OpenClaw Server → Groq

→ PlannedAction JSON (1 or more actions)

→ Android confirmation sheet

→ Samsung Calendar / Contacts / Notes

Privacy: Raw image never leaves the device. Only extracted text crosses the network



Impact & Use cases

Networking event: Snap a business card → contact saved + follow-up meeting booked in one tap

Conference / college fest: Snap the event poster → calendar event created with date, time, venue

Dinner / travel: Snap the receipt → expense logged instantly with merchant, amount, date

Team meeting: Snap the whiteboard → note created with all action items captured

Scale potential: Works on any Samsung Android device. The skill system is extensible — new action types (reminders, emails, URLs) can be added without changing the app. Every Samsung user who takes photos is a potential user.

Differentiation

Why it stands out: SnapTask is the only solution that goes from photo to executed action — not just recognized text.

MOAT 1 — Execution, not recognition: Every competitor (Lens, Live Text, Bixby) stops at showing you the text. SnapTask closes the loop by writing to Calendar, Contacts, and Notes automatically.

MOAT 2 — Multi-action intelligence: One image can produce multiple structured outputs simultaneously. A business card doesn't just save a contact — it books the follow-up. No other tool does this.

MOAT 3 — Privacy by architecture: The raw image never leaves the device. Only text is sent to the server. This is a structural guarantee, not a policy

Demo Link

GitHub Repo:

<https://github.com/AryanGupta21/SnapTask>

Demo Link:

<https://drive.google.com/drive/folders/1MtNhO165MN1cdbCr6dzKRSdJ2RapUnuv?usp=sharing>

Document submitted:

<https://drive.google.com/drive/folders/1MtNhO165MN1cdbCr6dzKRSdJ2RapUnuv?usp=sharing>

<Brownie Slide>

Title: What's next for SnapTask

More skills: Set reminders · Send emails · Open URLs · Scan Wi-Fi QR codes

Offline fallback: On-device Gemini Nano for when there's no network

Samsung DeX: Expanded confirmation UI on large screens

Bixby Routine integration: Auto-trigger SnapTask when camera roll detects new images

The skill architecture means any new integration is a single SKILL.md + index.js— the app never needs to change.